

Official-source Electricity compilation 05

Mixed-source chapter worksheet • cross-paper reuse disclosed

Total: 36 marks • Suggested time: 72 minutes

Instructions and reuse disclosure

Curated compilation of official CBSE source questions, not a CBSE-issued chapter paper. Original source labels and exact source images are retained.

Answer every selected question. Only one selected source-slot alternative appears in this paper; do not answer or count excluded alternatives.

Questions recur across this 15-paper collection. Reuse is disclosed per question in the source appendix; these are not 15 disjoint official papers.

Sections use available source marks: A MCQ; B 2 marks; C 3 marks; D 5 marks or source-marked case questions. Case marks are not inflated to five.

Section A — MCQ

Compilation questions 1-4; 4 marks. Exact source labels remain attached to each question.

Section B — 2-mark questions

Compilation questions 5-8; 8 marks. Exact source labels remain attached to each question.

Section C — 3-mark questions

Compilation questions 9-11; 9 marks. Exact source labels remain attached to each question.

Section D — 5-mark / case-based questions

Compilation questions 12-14; 15 marks. Exact source labels remain attached to each question.

Section A — MCQ

1. [1 marks] Source Q30 • 2026 First Board

30. The correct way to connect an ammeter and a voltmeter in an electric circuit is :

1

- (A) Ammeter in parallel and voltmeter in series
- (B) Ammeter and voltmeter both in parallel
- (C) Ammeter in series and voltmeter in parallel
- (D) Ammeter and voltmeter both in series

Source/editorial note: Visually confirmed complete replacement crop: question number, wording, all options/table values, mathematical symbols and printed marks. Whitespace normalized; mathematical glyphs represented in Unicode and stacked fractions as $\frac{1}{4}$ and $\frac{V}{4}$. Page/QR/footer debris removed from text; marks retained in marks field. No supplied diagram; table retained in text and evidence. Answers not reviewed. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

2. [1 marks] Source Q13 • 2022-23

13	A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire (a) decreases. (b) remains the same. (c) increases. (d) increases for some time and then decreases .	1
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3. [1 marks] Source Q14 • 2019-20

14 **Assertion:** A fuse wire is always connected in parallel with the mainline. 1

Reason: If a current larger than the specified value flows through the circuit, fuse wire melts.

For question numbers 13 and 14, two statements are given- one labeled *Assertion (A)* and the other labeled *Reason (R)*. Select the correct answer to these questions from the codes (i), (ii), (iii) and (iv) as given below

- i) Both A and R are true and R is correct explanation of the assertion.
- ii) Both A and R are true but R is not the correct explanation of the assertion.
- iii) A is true but R is false.
- iv) A is false but R is true.

13 **Assertion:** Following are the structural isomers of butane. 1

Source/editorial note: Fuse series placement and melting from excess current are explicitly NCERT Electricity section 11.7.1 (printed p190, PDF p20). Included after independent audit overturned the batch reviewer's domestic-circuit exclusion. Shared assertion-reason choices occur on preceding PDF page 5.

4. [1 marks] Source Q31 • 2026-27

31	An electrical appliance rated 1000 W is operated at 250 V. What current does it draw? A. 0.25 A B. 2.5 A C. 4.0 A D. 10 A	1
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Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section B — 2-mark questions

5. [2 marks] Source Q26 • main

- 26.** Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

2

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

6. [2 marks] Source Q25 main • 2023-24

25	A piece of wire of resistance R is cut into three equal parts. These parts are then connected in parallel. If the equivalent resistance of this parallel combination is R_1 , what is the value of the ratio $R_1 : R$?	2
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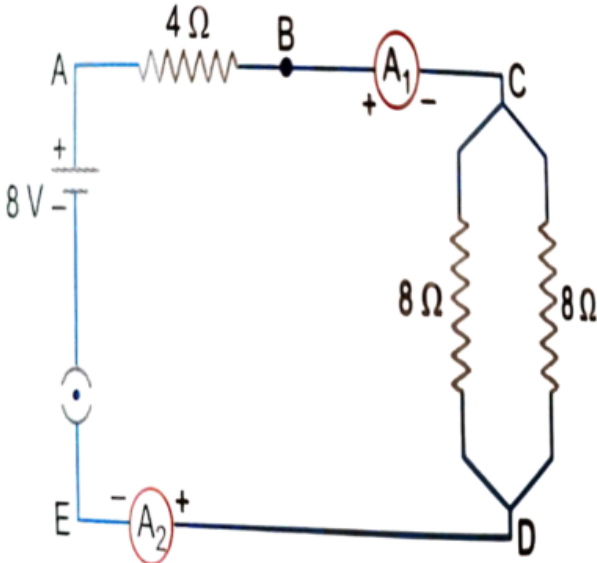
Source/editorial note: Main alternative only. OR concerns magnetic field patterns and is excluded. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

7. [2 marks] Source Q25 VI A • 2024-25

	<u>For visually impaired students</u>	
	A. We have four resistors A, B, C and D of resistance 3 Ω , 6 Ω , 9 Ω and 12 Ω respectively. Find out the lowest resistance which can be obtained by combining these four resistors.	

Source/editorial note: Official MS arithmetic error retained: $1/12 = 3/36$, not $2/36$. Independently calculated correct minimum = $36/25 = 1.44 \Omega$; this correction is not substituted into the official answer. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately. Consult the linked original marking scheme with this limitation in mind.

8. [2 marks] Source Q34 A • 2025-26

34	<u>Attempt either option A or B.</u> A.  Find out the following in the electric circuit given in the figure- (i) Effective resistance of two 8 ohm resistors in the combination. (ii) Current flowing through the 4-ohm resistor	2
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Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section C — 3-mark questions

9. [3 marks] Source Q27 • main

27. (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is $14\ \Omega$. If the resistivity of the material of the wire is $44 \times 10^{-8}\ \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)
- 3

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

10. [3 marks] Source Q31 • compartment

31. (a) Write mathematical expression for Joule's law of heating.
- (b) Calculate the amount of heat generated while transferring 72000 coulomb of charge in two hours through a potential difference of 220 V.
- 1+2

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

11. [3 marks] Source Q12 • 2018-19

12. Two lamps, one rated 100 W at 220 V and the other 200 W at 220V are connected (i) in series and (ii) in parallel to electric main supply of 220V. Find the current drawn in each case. 3

Source/editorial note: Included despite mains-supply context: self-contained resistance, power and current calculation, not domestic-wiring knowledge.

Section D — 5-mark / case-based questions

12. [5 marks] Source Q34 • main

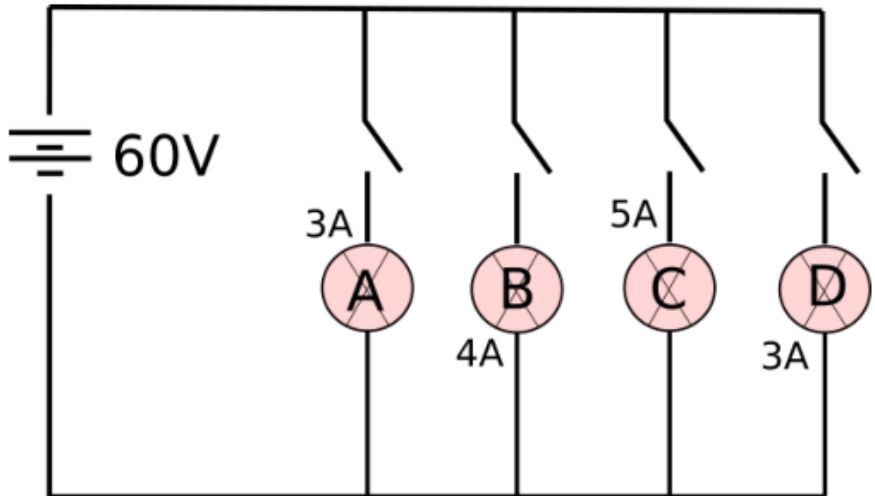
SECTION D

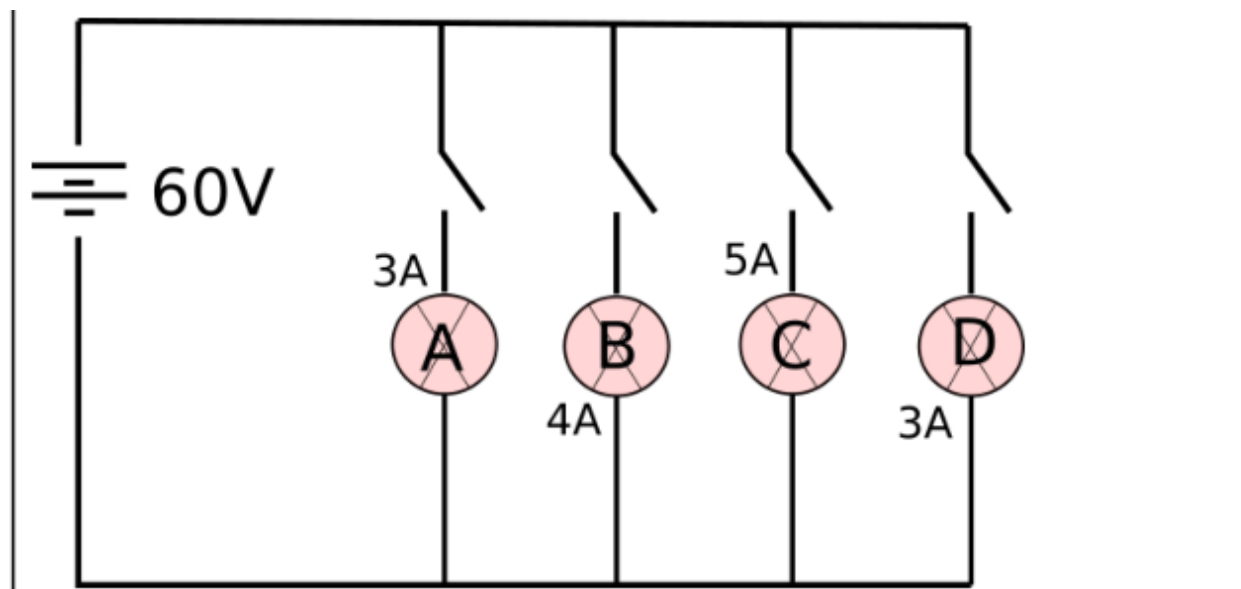
34. There are three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ joined in parallel in a circuit. The potential difference across the electric circuit is 10 V.

- (a) Draw a circuit diagram for the above case.
- (b) Find the total resistance of the combination of resistors.
- (c) Calculate the electric current drawn from the same source. 5

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

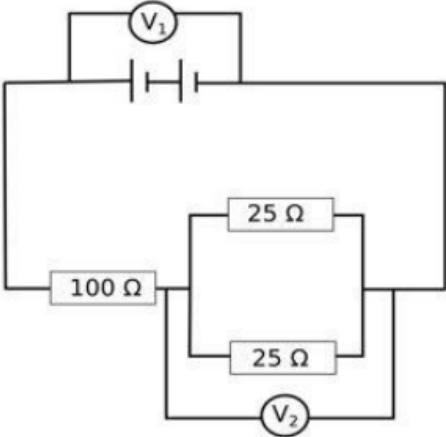
13. [5 marks] Source Q36 • 2020-21

36	In the given circuit, A, B, C and D are four lamps connected with a battery of 60V.  Analyse the circuit to answer the following questions. (i) What kind of combination are the lamps arranged in (series or parallel)? (ii) Explain with reference to your above answer, what are the advantages (any two) of this combination of lamps? (iii) Explain with proper calculations which lamp glows the brightest? (iv) Find out the total resistance of the circuit.	5
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Source/editorial note: All four subparts retained. The OR alternative on pp.14-15 concerns magnetic fields and the right-hand thumb rule and is excluded. Source MS p.8 prints " $= 60/15 \text{ A} = 4\text{A}$ "; the resistance unit is corrected explicitly to Ω , not silently attributed to the source.

14. [5 marks] Source Q39 A • 2025-26

39	<u>Attempt either option A or B.</u> A.  The arrangement of resistors shown in the above figure is connected to a battery.; The power dissipation in the $100\ \Omega$ resistor is $81\ \text{W}$. Calculate - the current in the circuit - the reading in the voltmeter V_2 - the reading in the voltmeter V_1	5
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Source/editorial note: Original “battery.,” retained. Total 5 marks; MS provides no subpart allocation. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — pyq-2026-0c419e09ddf75b6dd3da937f

Official answer unavailable/withheld: No independently verified official answer supplied.

Original marking scheme (see limitation above): https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086_SCIENCE.zip

Question 2 — sqp_2022-23_q13

(c) increases

13.	(c) increases	1
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

Question 3 — SQP-2019-20-Q14

iv) Assertion is false but reason is true.

14	iv) Assertion is false but reason is true.	1 mark
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Marking-scheme PDF page(s): [2]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

Question 4 — SQP_2026-27_q31

C. 4.0 A

31	C. 4.0 A	1
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Marking-scheme PDF page(s): [7]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-MS.pdf

Question 5 — pyq-2024-1e2d386b35acad1e9e07f369

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 6 — SQP_2023-24_q25_main

Resistance of each part is $R/3 \Omega$ (as resistance is proportional to the length of the wire.) [0.5]

$$1/R_1 = 3/R + 3/R + 3/R = 9/R \quad [1]$$

$$\therefore R_1 = R/9 \therefore R_1/R = 1/9 \quad [0.5]$$

25	Resistance of each part is $\frac{R}{3} \Omega$ (as resistance is proportional to the length of the wire.) $\frac{1}{R_1} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$ $\therefore R_1 = \frac{R}{9} \therefore \frac{R_1}{R} = \frac{1}{9}$ OR	2 [0.5] [1] [0.5]
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Marking-scheme PDF page(s): [2]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

Question 7 — SQP_2024-25_q25_VI_A

Official answer unavailable/withheld: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately.

Original marking scheme (see limitation above): https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-M S.pdf

Question 8 — SQP_2025-26_q34_A

A.

(i) $R = R_1 R_2 / (R_1 + R_2) = (8 \times 8) / (8 + 8) = 4 \text{ ohms}$

(ii) $I = V/R = 8/(4+4) = 1 \text{ A}$

34	<u>Student to attempt either A or B.</u>	2
	A.	
	(i)	
	$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{8 \times 8}{8 + 8} = 4 \text{ ohms}$	
	(ii)	
	$I = \frac{V}{R} = \frac{8}{(4 + 4)} = 1 \text{ A}$	
	OR	

Marking-scheme PDF page(s): [7]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

Question 9 — pyq-2025-74354640f102dee3b9237ff5

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 10 — pyq-2025-179b945e5fe9505f3715ef55

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 11 — cbse_sqp_2018-19_electricity_q12

$R_1 = V^2/P_1 = 484 \Omega$; $R_2 = V^2/P_2 = 242 \Omega$. In series: $R_s = 726 \Omega$, $I_s = 220/726 = 10/33 \text{ A} = 0.30 \text{ A}$. In parallel: $R_p = 484/3 \Omega$, $I_p = 220 \times 3/484 = 30/22 \text{ A} = 1.36 \text{ A}$.

$$12. R_1 = \frac{V^2}{P_1} = \frac{220 \times 220}{100} = 484 \, \Omega$$

$$R_2 = \frac{V^2}{P_2} = \frac{220 \times 220}{200} = 242 \, \Omega$$

In series:

$$R_s = R_1 + R_2 = 484 + 242 = 726 \, \Omega$$

$$\therefore I_s = \frac{V}{R_s} = \frac{220}{726} = \frac{10}{33} \, \text{A} = 0.30 \, \text{A}$$

In parallel:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{1}{484} + \frac{1}{242}$$

$$R_p = \frac{484}{3} \, \Omega$$

$$I_p = \frac{V}{R_p} = \frac{220 \times 3}{484} = \frac{30}{22} \, \text{A} = 1.36 \, \text{A}$$

Marking-scheme PDF page(s): [4]

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_MS_2018-19.pdf

Question 12 — pyq-2023-84af9d4edc02297aecab4a0f

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 13 — sqp_2020-21_q36_main

(i) The lamps are in parallel. (ii) A faulty lamp does not affect the working of the others; all lamps use the full potential of the battery. (iii) The lamp with the highest power glows brightest. All have the same voltage; C has the largest current. $P_C = 5 \times 60 = 300 \, \text{W}$, the maximum. (iv) Total current = $3 + 4 + 5 + 3 = 15 \, \text{A}$; $R = V/I = 60/15 = 4 \, \Omega$. The MS prints the final unit incorrectly as A.

36	The lamps are in parallel.	1
	Advantages:	
	If one lamp is faulty, it will not affect the working of the other lamps. They will also be using the full potential of the battery as they are connected in parallel.	1
	The lamp with the highest power will glow the brightest.	1
	$P=VI$	
	In this case, all the bulbs have the same voltage. But lamp C has the highest current.	1
	Hence, for Lamp C $P=5 \times 60 \, \text{Watt} = 300 \, \text{W}$. (the maximum).	
	The total current in the circuit = $3+4+5+3 \, \text{A} = 15\text{A}$	1
	The Voltage = 60V	
	$V=IR$ and hence $R = V/I$	
	$= 60/15 \, \text{A} = 4\text{A}$	

Marking-scheme PDF page(s): [8]

Question 14 — SQP_2025-26_q39_A

A.

(i) Power across the $100\ \Omega$ resistance = $81\ \text{W}$

$$P = I^2 R = 81\ \text{W}$$

$$\therefore I^2 = 81/100$$

$$\therefore I = \sqrt{(81/100)} = 9/10 = 0.9\ \text{A}$$

(ii) Voltage across the $25\ \Omega$ resistors = $V_2 = IR_{\text{eqv}}$ for the $25\ \Omega$ resistors

$$1/R_{\text{eqv}} = 1/25 + 1/25 = 2/25$$

$$\therefore R_{\text{eqv}} = 25/2 = 12.5\ \Omega$$

$$\therefore V_2 = 0.9\ \text{A} \times 12.5\ \Omega = 11.25\ \text{V}$$

(iii) Voltage across $100\ \Omega = V_{100} = IR = 0.9\ \text{A} \times 100\ \Omega = 90\ \text{V}$

$$\therefore V_1 = 90\ \text{V} + 11.25\ \text{V} = 101.25\ \text{V}$$

39

Students to attempt either option A or B.

5

A.

(i) Power across the $100\ \Omega$ resistance = $81\ \text{W}$

$$P = I^2 R = 81\ \text{W}$$

$$\therefore I^2 = \frac{81}{100}$$

$$\therefore I = \sqrt{\frac{81}{100}} = \frac{9}{10} = 0.9\ \text{A}$$

(ii) Voltage across the $25\ \Omega$ resistors = $V_2 = IR_{\text{eqv}}$
for the $25\ \Omega$ resistors

$$\frac{1}{R_{\text{eqv}}} = \frac{1}{25} + \frac{1}{25} = \frac{2}{25}$$

$$\therefore R_{\text{eqv}} = \frac{25}{2} = 12.5\ \Omega$$

$$\therefore V_2 = 0.9\ \text{A} \times 12.5\ \Omega = 11.25\ \text{V}$$

(iii) Voltage across $100\ \Omega = V_{100} = IR = 0.9\ \text{A} \times 100\ \Omega = 90\ \text{V}$

$$\therefore V_1 = 90\ \text{V} + 11.25\ \text{V} = 101.25\ \text{V}$$

OR

Marking-scheme PDF page(s): [12, 13]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

Per-question source appendix

Original CBSE questions/images remain attributed to their source. Compilation selection and numbering are editorial. The full Science source PDFs may include unrelated chapters; only selected Electricity crops form this worksheet.

1. pyq-2026-0c419e09ddf75b6dd3da937f

CBSE Class X Science 2026 First Board 31-3-3 | PYQ | Year/session: 2026 / 2026 First Board | Paper code: 31/3/3 | Source question: 30 | Original section: Section C (Physics) | Selected marks: 1 | PDF pages (one-based): 23

<https://www.cbse.gov.in/cbsenew/question-paper/2026/X/Science.zip>

Also used in: official-13

2. sqp_2022-23_q13

CBSE Class X Science (086) Sample Question Paper 2022-23 | SQP | Year/session: 2022-23 / 2022-23 | Paper code: Not recorded | Source question: 13 | Original section: Section A | Selected marks: 1 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Also used in: official-03, official-06, official-09, official-13, official-15

3. SQP-2019-20-Q14

Science-086 Sample Question Paper 2019-20 Class-X | SQP | Year/session: 2019-20 / 2019-20 | Paper code: Not recorded | Source question: 14 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 5, 6

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Also used in: official-03, official-09, official-11, official-13

4. SQP_2026-27_q31

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 31 | Original section: Section - C | Selected marks: 1 | PDF pages (one-based): 11

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Also used in: official-01, official-08, official-11, official-12, official-15

5. pyq-2024-1e2d386b35acad1e9e07f369

CBSE — Class X Science — 2024 — main — 31/4/1 | PYQ | Year/session: 2024 / main | Paper code: 31/4/1 | Source question: 26 | Original section: SECTION B | Selected marks: 2 | PDF pages (one-based): 13

<https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>

Also used in: official-08, official-11, official-14

6. SQP_2023-24_q25_main

CBSE Class X Science (086) Sample Question Paper 2023-24 (English) | SQP | Year/session: 2023-24 / 2023-24 | Paper code: Not recorded | Source question: 25 main | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Also used in: official-06

7. SQP_2024-25_q25_VI_A

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 25 VI A | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 7

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Also used in: official-06, official-12

8. SQP_2025-26_q34_A

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 34 A | Original section: Section – C | Selected marks: 2 | PDF pages (one-based): 11

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Also used in: official-08, official-12

9. pyq-2025-74354640f102dee3b9237ff5

CBSE — Class X Science — 2025 — main — 31/1/2 | PYQ | Year/session: 2025 / main | Paper code: 31/1/2 | Source question: 27 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 15

https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip

Also used in: official-08, official-15

10. pyq-2025-179b945e5fe9505f3715ef55

CBSE — Class X Science — 2025 — compartment — 31/S/3 | PYQ | Year/session: 2025 / compartment | Paper code: 31/S/3 | Source question: 31 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 15

<https://www.cbse.gov.in/cbsenew/question-paper/2025-COMPTT/X/Science.zip>

Also used in: official-02, official-11

11. cbse_sqp_2018-19_electricity_q12

Class-X Science-086 SAMPLE QUESTION PAPER-19 | SQP | Year/session: 2018-19 / 2018-19 | Paper code: Not recorded | Source question: 12 | Original section: SECTION C | Selected marks: 3 | PDF pages (one-based): 2

https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

Also used in: official-01

12. pyq-2023-84af9d4edc02297aecab4a0f

CBSE — Class X Science — 2023 — main — 31/5/2 | PYQ | Year/session: 2023 / main | Paper code: 31/5/2 | Source question: 34 | Original section: SECTION D | Selected marks: 5 | PDF pages (one-based): 19

<https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>

Also used in: official-01, official-09, official-10, official-13

13. sqp_2020-21_q36_main

CBSE Class X Science (086) Sample Question Paper 2020-21 | SQP | Year/session: 2020-21 / 2020-21 | Paper code: Not recorded | Source question: 36 | Original section: Section D | Selected marks: 5 | PDF pages (one-based): 14

https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf

Also used in: official-03, official-08, official-12

14. SQP_2025-26_q39_A

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 39 A | Original section: Section – C | Selected marks: 5 | PDF pages (one-based): 14

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

No cross-paper reuse.